

Mohamed Shamik

Software Engineering Undergraduate

mdshamik@outlook.com | +65 8262 3134 | GitHub: <https://github.com/Mohdshamik11> | LinkedIn: <https://www.linkedin.com/in/mohamedshamik/>

SUMMARY

Software Engineering undergraduate with hands-on experience across full-stack development and IT infrastructure, gained through internships and academic projects. Proficient in Python, SQL, Java and cloud/DevOps tooling (Docker, AWS, CI/CD). Actively learning and building meaningful software with AI and automation integration.

SKILLS

Languages: Python, SQL, Java
Frameworks: PyTorch, FastAPI, Flask
Cloud & DevOps: AWS (EC2), Vercel, GitHub Actions (CI/CD), Nginx
Tools & Platforms: Docker, GitHub, VSCode
Databases: Supabase, MySQL

EDUCATION

B.Eng in Software Engineering (Hons) Singapore Institute of Technology	Sep 2024 – Ongoing
Diploma in Infocomm & Security Nanyang Polytechnic	Apr 2019 – May 2022

PROFESSIONAL EXPERIENCE

Micro2000 IT Installer	May 2025 – Aug 2025
- Executed device provisioning, sanitisation, and assessment for the Schools Standard ICT Operating Environment (SSOE) project. - Managed small team operations and liaised with school staff during handovers and takeovers. - Independently handled devices, documentation, and on-site project tasks, demonstrating accountability. - Contributed to a smooth SSOE rollout, ensuring 100% compliance with project standards.	
Zicom Holdings Pte Ltd IT & Web Development Intern	Dec 2021 – Apr 2022
- Revamped the company's official website using a CMS, updating product and service listings to keep content current and improve customer-facing presentation. - Consulted with the IT Manager to assess logistics and hardware upgrade needs, coordinating procurement of necessary equipment. - Delivered IT Helpdesk support across hardware, software, and network connectivity issues for the IT and Engineering departments. - Conducted a virtual training session for a team of 5, sharing networking fundamentals to strengthen the team's technical knowledge base.	

ACADEMIC PROJECTS

Bachelor of Engineering in Software Engineering - Singapore Institute of Technology

1. HitsV2 - Ongoing
 - Building a full-stack web portal (React, FastAPI, Databricks) that lets users discover, preview, and request access to smart-campus IoT datasets across 4 domains.
 - Designed an access-request workflow where dataset owners approve or revoke access, with notifications to keep users and owners informed.
 - Implemented secure login and role-based permissions (user, owner, admin, custodian) so users only see and download data they are authorised to access.
 - Architected the system so the underlying data storage can be upgraded later without disrupting the user-facing application.
2. Full Stack Application with Security Measures
 - Built a web app for personal finance tracking, with consent-based data sharing between individuals, households, and financial advisors.
 - Set up an automated pipeline that tests, checks, and deploys the app to the cloud every time code changes are merged.
 - Protected user accounts with secure login and safeguards against repeated unauthorised login attempts.
 - Designed an email invitation system that lets users bring household members and advisors onto the app with appropriate, limited access.
3. Assisted Remote-Controlled Ball Rover
 - Built an end-to-end IoT system for a remote-controlled rover, combining on-board device software with a browser dashboard for real-time control.
 - Added automatic obstacle detection using multiple sensors, with a safety override that takes over from manual control to prevent collisions.
 - Developed a 360° autonomous scanning feature that surveys the rover's surroundings and shows a live map of nearby obstacles in the web interface.

Diploma in Infocomm & Security - Nanyang Polytechnic

1. Network Design and Implementation
 - Designed and configured a multi-router, multi-switch network topology, applying subnetting and routing protocols to ensure reliable connectivity across segments.
 - Implemented network security controls including firewalls, Intrusion Prevention Systems (IPS), VPN tunnels, and traffic encryption to safeguard against unauthorized access.
 - Configured user authentication and role-based access controls, restricting server access to authorized personnel only.

PERSONAL PROJECTS

1. Image Quality Classifier & Enhancer
 - Built an AI tool that automatically detects five common photo problems in one pass: blur, over-exposure, under-exposure, noise, and low contrast.
 - Created the training data from scratch by deliberately degrading clean photos, with labels generated automatically, so no manual annotation was needed.
 - Developed a second AI model that repairs blurry images and produces genuinely sharp results, and used standard image enhancement techniques to fix exposure, contrast, and noise.
 - Deployed the tool as a live web app where users upload a photo and get back the detected issues along with an enhanced version.

CERTIFICATIONS

- Professional Scrum Master I (PSM I) - [Scrum.org](https://www.scrum.org), Sep 2026
- AWS Certified Cloud Practitioner — In Progress